

Apex Diagnostics

Comprehensive Pre-Marital Health Report

Clinical Profile: Male Hypercholesterolemia

Name: Mr. Manish Patel

Age/Sex: 24 Yrs / M

Date: 02 Jun 2026, 03:50 PM

ID: ADX-38923

| BLOOD GROUP - ABO TYPING     |        |           |      |
|------------------------------|--------|-----------|------|
| Parameter                    | Result | Reference | Unit |
| Blood Group (A / B / AB / O) | O      | -         | -    |

| BLOOD GROUP - RH (RHESUS) TYPING |          |           |      |
|----------------------------------|----------|-----------|------|
| Parameter                        | Result   | Reference | Unit |
| Rh Factor (Positive / Negative)  | Positive | -         | -    |

| COMPLETE BLOOD COUNT (CBC) / HEMOGRAM             |        |            |         |
|---------------------------------------------------|--------|------------|---------|
| Parameter                                         | Result | Reference  | Unit    |
| Total Red Blood Cell Count (RBC)                  | 5.0    | 4.5-5.9    | mill/μL |
| Hemoglobin (Hb)                                   | 14.8   | 13.5-17.5  | g/dL    |
| Hematocrit / Packed Cell Volume (PCV)             | 43.1   | 41.0-53.0  | %       |
| Mean Corpuscular Volume (MCV)                     | 86.3   | 80.0-100.0 | fL      |
| Mean Corpuscular Hemoglobin (MCH)                 | 31.8   | 27.0-33.0  | pg      |
| Mean Corpuscular Hemoglobin Concentration (MCHC)  | 34.3   | 32.0-36.0  | g/dL    |
| Red Cell Distribution Width - CV (RDW-CV)         | 13.4   | 11.5-14.5  | %       |
| Red Cell Distribution Width - SD (RDW-SD)         | 41.3   | 39.0-46.0  | fL      |
| Total White Blood Cell Count (WBC / TLC)          | 7.0    | 4.5-11.0   | thou/μL |
| Neutrophils (%)                                   | 55.4   | 40-80      | %       |
| Neutrophils - Absolute Count                      | 3.88   | 2.0-7.0    | thou/μL |
| Lymphocytes (%)                                   | 25.9   | 20-40      | %       |
| Lymphocytes - Absolute Count                      | 2.1    | 1.0-3.0    | thou/μL |
| Monocytes (%)                                     | 5.5    | 2-10       | %       |
| Monocytes - Absolute Count                        | 0.42   | 0.2-1.0    | thou/μL |
| Eosinophils (%)                                   | 1.6    | 1-6        | %       |
| Eosinophils - Absolute Count                      | 0.11   | 0.02-0.5   | thou/μL |
| Basophils (%)                                     | 0.5    | 0-2        | %       |
| Basophils - Absolute Count                        | 0.04   | 0.0-0.1    | thou/μL |
| Immature Granulocytes (%)                         | 0.2    | 0-0.5      | %       |
| Immature Granulocytes - Absolute Count            | 0.01   | 0-0.03     | thou/μL |
| Nucleated Red Blood Cells (NRBC) - Absolute Count | 0.0    | 0.0        | thou/μL |
| Nucleated Red Blood Cells (NRBC) %                | 0.0    | 0.0        | %       |
| Platelet Count                                    | 273    | 150-450    | thou/μL |
| Mean Platelet Volume (MPV)                        | 9.5    | 7.5-11.5   | fL      |
| Platelet Distribution Width (PDW)                 | 12.0   | 9.0-17.0   | fL      |
| Plateletcrit (PCT)                                | 0.25   | 0.15-0.35  | %       |
| Platelet-Large Cell Ratio (P-LCR)                 | 25.0   | 15.0-35.0  | %       |

| ERYTHROCYTE SEDIMENTATION RATE (ESR) |        |           |       |
|--------------------------------------|--------|-----------|-------|
| Parameter                            | Result | Reference | Unit  |
| Erythrocyte Sedimentation Rate (ESR) | 10     | < 15      | mm/hr |

| BLOOD SUGAR - FASTING (FBS / FBG) |        |           |       |
|-----------------------------------|--------|-----------|-------|
| Parameter                         | Result | Reference | Unit  |
| Fasting Blood Glucose (FBG)       | 88.0   | 70-100    | mg/dL |

| BLOOD SUGAR - RANDOM (RBS / BSR) |        |           |       |
|----------------------------------|--------|-----------|-------|
| Parameter                        | Result | Reference | Unit  |
| Random Blood Sugar (RBS)         | 103.6  | 70-140    | mg/dL |

| GLYCATED HEMOGLOBIN (HBA1C)     |        |           |       |
|---------------------------------|--------|-----------|-------|
| Parameter                       | Result | Reference | Unit  |
| HbA1c (%)                       | 5.1    | 4.0-5.6   | %     |
| Estimated Average Glucose (eAG) | 99.7   | 70-126    | mg/dL |

| AVERAGE BLOOD GLUCOSE (ABG) |        |           |       |
|-----------------------------|--------|-----------|-------|
| Parameter                   | Result | Reference | Unit  |
| Average Blood Glucose (ABG) | 103.0  | < 117     | mg/dL |

| LIVER FUNCTION TEST (LFT) - TOTAL BILIRUBIN |        |           |       |
|---------------------------------------------|--------|-----------|-------|
| Parameter                                   | Result | Reference | Unit  |
| Total Bilirubin                             | 0.82   | 0.3-1.2   | mg/dL |
| Direct (Conjugated) Bilirubin               | 0.2    | < 0.3     | mg/dL |
| Indirect (Unconjugated) Bilirubin           | 0.2    | < 0.3     | mg/dL |

| LIVER FUNCTION TEST (LFT) - SGPT / ALT |        |           |      |
|----------------------------------------|--------|-----------|------|
| Parameter                              | Result | Reference | Unit |
| Alanine Aminotransferase (SGPT / ALT)  | 19     | < 41      | U/L  |

| LIVER FUNCTION TEST (LFT) - SGOT / AST  |        |           |      |
|-----------------------------------------|--------|-----------|------|
| Parameter                               | Result | Reference | Unit |
| Aspartate Aminotransferase (SGOT / AST) | 21     | < 40      | U/L  |

| KIDNEY FUNCTION TEST (KFT)                         |        |           |        |
|----------------------------------------------------|--------|-----------|--------|
| Parameter                                          | Result | Reference | Unit   |
| Serum Creatinine                                   | 0.91   | 0.6-1.2   | mg/dL  |
| Blood Urea Nitrogen (BUN)                          | 15     | 7-20      | mg/dL  |
| BUN / Creatinine Ratio                             | 16.5   | 10-20     | Ratio  |
| Serum Uric Acid                                    | 4.1    | 3.5-7.2   | mg/dL  |
| Serum Urea                                         | 32     | 15-40     | mg/dL  |
| Estimated Glomerular Filtration Rate (eGFR)        | 104    | > 90      | mL/min |
| Serum Sodium (Na <sup>+</sup> )                    | 140    | 135-145   | mEq/L  |
| Serum Potassium (K <sup>+</sup> )                  | 4.2    | 3.5-5.1   | mEq/L  |
| Serum Chloride (Cl <sup>-</sup> )                  | 102    | 98-107    | mEq/L  |
| Serum Bicarbonate (HCO <sub>3</sub> <sup>-</sup> ) | 26     | 22-29     | mEq/L  |

| SERUM IRON |        |           |       |
|------------|--------|-----------|-------|
| Parameter  | Result | Reference | Unit  |
| Serum Iron | 95     | 50-170    | μg/dL |

| TOTAL IRON BINDING CAPACITY (TIBC) |        |           |       |
|------------------------------------|--------|-----------|-------|
| Parameter                          | Result | Reference | Unit  |
| Total Iron Binding Capacity (TIBC) | 302    | 250-450   | μg/dL |

| UNSATURATED IRON BINDING CAPACITY (UIBC) |        |           |       |
|------------------------------------------|--------|-----------|-------|
| Parameter                                | Result | Reference | Unit  |
| Unsaturated Iron Binding Capacity (UIBC) | 207    | 110-370   | μg/dL |

| TRANSFERRIN SATURATION (%) |        |           |      |
|----------------------------|--------|-----------|------|
| Parameter                  | Result | Reference | Unit |
| Transferrin Saturation (%) | 31     | 20-50     | %    |

| SERUM FERRITIN |        |           |       |
|----------------|--------|-----------|-------|
| Parameter      | Result | Reference | Unit  |
| Serum Ferritin | 117    | 20-250    | ng/mL |

| HEMOGLOBIN HPLC / VARIANT ANALYSIS    |              |              |      |
|---------------------------------------|--------------|--------------|------|
| Parameter                             | Result       | Reference    | Unit |
| Hemoglobin A (HbA)                    | 97.2         | > 95.0       | %    |
| Hemoglobin A2 (HbA2)                  | 2.6          | 1.5-3.5      | %    |
| Hemoglobin F (HbF / Fetal Hemoglobin) | 0.5          | < 2.0        | %    |
| Hemoglobin S (HbS)                    | 0.0          | 0.0          | %    |
| Hemoglobin C (HbC)                    | 0.0          | 0.0          | %    |
| Hemoglobin D (HbD)                    | 0.0          | 0.0          | %    |
| Hemoglobin E (HbE)                    | 0.0          | 0.0          | %    |
| Total Hemoglobin (g/dL)               | 14.8         | 12.0-16.0    | g/dL |
| Unknown / Other Hemoglobin Fractions  | 0.0          | 0.0          | %    |
| Interpretation / Diagnosis            | Normal Study | Normal Study | -    |

| HEMOGLOBIN A2          |        |           |      |
|------------------------|--------|-----------|------|
| Parameter              | Result | Reference | Unit |
| Hemoglobin A2 (HbA2) % | 2.6    | 1.5-3.5   | %    |

| HEMOGLOBIN C         |        |           |      |
|----------------------|--------|-----------|------|
| Parameter            | Result | Reference | Unit |
| Hemoglobin C (HbC) % | 0.0    | 0.0       | %    |

| HEMOGLOBIN D         |        |           |      |
|----------------------|--------|-----------|------|
| Parameter            | Result | Reference | Unit |
| Hemoglobin D (HbD) % | 0.0    | 0.0       | %    |

| HEMOGLOBIN F                            |        |           |      |
|-----------------------------------------|--------|-----------|------|
| Parameter                               | Result | Reference | Unit |
| Hemoglobin F / Fetal Hemoglobin (HbF) % | 0.5    | < 2.0     | %    |

| HEMOGLOBIN S         |        |           |      |
|----------------------|--------|-----------|------|
| Parameter            | Result | Reference | Unit |
| Hemoglobin S (HbS) % | 0.0    | 0.0       | %    |

| THYROID STIMULATING HORMONE (TSH)        |        |           |        |
|------------------------------------------|--------|-----------|--------|
| Parameter                                | Result | Reference | Unit   |
| Thyroid Stimulating Hormone (TSH / uTSH) | 1.52   | 0.4-4.0   | µIU/mL |
| Triiodothyronine (T3) - Total            | 120    | 80-200    | ng/dL  |
| Thyroxine (T4) - Total                   | 8.0    | 5.0-12.0  | µg/dL  |
| Free T3 (FT3)                            | 3.3    | 2.3-4.2   | pg/mL  |
| Free T4 (FT4)                            | 1.2    | 0.9-1.7   | ng/dL  |

| VITAMIN B12 (CYANOCOBALAMIN)    |        |           |       |
|---------------------------------|--------|-----------|-------|
| Parameter                       | Result | Reference | Unit  |
| Vitamin B12 (Cobalamin) - Serum | 427    | 200-900   | pg/mL |

| LIPID PROFILE / LIPID SCREEN                 |        |           |       |
|----------------------------------------------|--------|-----------|-------|
| Parameter                                    | Result | Reference | Unit  |
| Total Cholesterol                            | 288    | < 200     | mg/dL |
| High-Density Lipoprotein Cholesterol (HDL-C) | 51     | > 40      | mg/dL |
| Low-Density Lipoprotein Cholesterol (LDL-C)  | 194    | < 100     | mg/dL |
| Very Low-Density Lipoprotein (VLDL-C)        | 18     | 2-30      | mg/dL |
| Triglycerides                                | 109    | < 150     | mg/dL |
| Non-HDL Cholesterol                          | 237    | < 130     | mg/dL |
| Total Cholesterol / HDL Ratio (TC:HDL)       | 5.6    | < 5.0     | Ratio |
| LDL / HDL Ratio                              | 3.8    | < 3.0     | Ratio |

| SYPHILIS - VDRL / RPR                       |                |                |      |
|---------------------------------------------|----------------|----------------|------|
| Parameter                                   | Result         | Reference      | Unit |
| VDRL / RPR Result (Reactive / Non-Reactive) | Non-Reactive   | Non-Reactive   | -    |
| RPR Titre (if reactive)                     | Not Applicable | Not Applicable | -    |

| HEPATITIS B SURFACE ANTIGEN (HBSAG)                  |              |              |      |
|------------------------------------------------------|--------------|--------------|------|
| Parameter                                            | Result       | Reference    | Unit |
| HBsAg - Qualitative Result (Reactive / Non-Reactive) | Non-Reactive | Non-Reactive | -    |
| HBsAg - OD Value / Signal-to-Cutoff Ratio            | 0.2          | < 0.9        | S/CO |

| HEPATITIS C VIRUS ANTIBODY (ANTI-HCV)                            |              |              |      |
|------------------------------------------------------------------|--------------|--------------|------|
| Parameter                                                        | Result       | Reference    | Unit |
| Anti-HCV Antibody - Qualitative Result (Reactive / Non-Reactive) | Non-Reactive | Non-Reactive | -    |
| Anti-HCV - OD Value / Signal-to-Cutoff Ratio                     | 0.1          | < 0.9        | S/CO |

| HIV 1 & 2 ANTIBODY TEST                               |              |              |      |
|-------------------------------------------------------|--------------|--------------|------|
| Parameter                                             | Result       | Reference    | Unit |
| HIV 1 & 2 Antibody - Result (Reactive / Non-Reactive) | Non-Reactive | Non-Reactive | -    |
| HIV ICT (Immunochromatographic Test) Result           | Non-Reactive | Non-Reactive | -    |

| HIV COMBO (P24 ANTIGEN + ANTIBODY) TEST               |              |              |      |
|-------------------------------------------------------|--------------|--------------|------|
| Parameter                                             | Result       | Reference    | Unit |
| HIV P24 Antigen - Result (Detected / Not Detected)    | Not Detected | Not Detected | -    |
| HIV 1 & 2 Antibody - Result (Reactive / Non-Reactive) | Non-Reactive | Non-Reactive | -    |
| HIV Combo Assay - OD Value / S/CO Ratio               | 0.1          | < 0.9        | S/CO |
| HIV 1 Differentiation Result                          | Negative     | Negative     | -    |
| HIV 2 Differentiation Result                          | Negative     | Negative     | -    |

| URINE ROUTINE & MICROSCOPY EXAMINATION |             |             |         |
|----------------------------------------|-------------|-------------|---------|
| Parameter                              | Result      | Reference   | Unit    |
| Colour                                 | Pale Yellow | Pale Yellow | -       |
| Appearance / Clarity / Turbidity       | Clear       | Clear       | -       |
| Specific Gravity                       | 1.015       | 1.005-1.030 | -       |
| pH                                     | 6.0         | 4.5-8.0     | -       |
| Urinary Protein                        | Absent      | Absent      | -       |
| Urinary Glucose                        | Absent      | Absent      | -       |
| Urinary Ketones                        | Absent      | Absent      | -       |
| Urinary Bilirubin                      | Absent      | Absent      | -       |
| Urobilinogen                           | Normal      | Normal      | -       |
| Blood / Haemoglobin in Urine           | Absent      | Absent      | -       |
| Nitrites                               | Absent      | Absent      | -       |
| Leukocyte Esterase                     | Absent      | Absent      | -       |
| Pus Cells (WBCs) per HPF               | 1-2         | 0-5         | per HPF |
| Red Blood Cells (RBCs) per HPF         | 0-1         | 0-2         | per HPF |
| Epithelial Cells                       | Rare        | Rare        | per HPF |
| Casts                                  | 21          | < 40        | U/L     |
| Crystals                               | Absent      | Absent      | -       |
| Bacteria                               | Absent      | Absent      | -       |
| Yeast / Fungi                          | 21          | < 40        | U/L     |
| Mucus Threads                          | Absent      | Absent      | -       |
| Amorphous Material                     | 6.0         | 4.5-8.0     | -       |

**SEMEN ANALYSIS**

| Parameter                            | Result          | Reference       | Unit           |
|--------------------------------------|-----------------|-----------------|----------------|
| Semen Volume                         | 3.2             | >= 1.5          | mL             |
| Semen pH                             | 7.5             | >= 7.2          | -              |
| Colour / Appearance                  | Pale Yellow     | Pale Yellow     | -              |
| Viscosity                            | Normal          | Normal          | -              |
| Liquefaction Time                    | 25              | < 60            | mins           |
| Sperm Concentration                  | 54              | >= 15           | mill/mL        |
| Total Sperm Count per Ejaculate      | 173             | >= 39           | mill/<br>ejac. |
| Total Sperm Motility (%)             | 61              | >= 42           | %              |
| Progressive Motility (PR) %          | 46              | >= 30           | %              |
| Non-Progressive Motility (NP) %      | 46              | >= 30           | %              |
| Immotility (IM) %                    | 39              | -               | %              |
| Total Motile Sperm Count (TMSC)      | 105             | -               | million        |
| Sperm Morphology - Normal Forms (%)  | 6.0             | 4.5-8.0         | -              |
| Sperm Head Defects (%)               | 60              | -               | %              |
| Sperm Midpiece Defects (%)           | 20              | -               | %              |
| Sperm Tail Defects (%)               | 11              | -               | %              |
| Sperm Vitality / Viability (%)       | 75              | >= 54           | %              |
| Pus Cells / Leucocytes (per HPF)     | 1-2             | 0-5             | per HPF        |
| Round Cells per mL                   | < 1             | < 1             | mill/mL        |
| Sperm Agglutination                  | Absent          | Absent          | -              |
| Interpretation / Fertility Prognosis | Normozoospermia | Normozoospermia | -              |